

The source or cause of the Sun's light, and the production of all elements in the periodic table, is rooted in nuclear physics and stellar evolution.

Light is created in the Sun by the nuclear fusion process in its core. It is here that enormous pressure and temperatures in excess of 15 million Kelvin force hydrogen nuclei together to form helium. This reaction is governed by the equation which everyone has heard of: $E = mc^2$. In this equation, the small amount of mass lost during the reaction is transformed into an enormous amount of energy in the form of photons.

Although the Big Bang created the universe's hydrogen and helium, the rest of the periodic table was formed in stars in a process called nucleosynthesis. Stars like our Sun will fuse up to the point of helium but bigger stars go further up the periodic table fusing carbon, oxygen, and neon etc. In a series of shells. This process continues until iron has been formed in the core of the star and the core has no more energy available in this fusion process as it requires more energy to produce iron than it yields. This collapse and explosion, called a supernova then gives the temperatures and pressures needed to create elements higher than iron, such as gold and uranium. This matter is then dispersed through the galaxy to seed later generations of stars and planets.

The general process by which a large diffuse cloud of gas turns into a star and surrounding planets is known as the nebular hypothesis.

The process started in a vast molecular cloud-a cold, diffused body of hydrogen and interstellar dust. If a part of it reaches a certain density, or gets hit by a shockwave, gravity makes it collapse. As it collapses, it starts to spin faster due to the conservation of angular momentum, the same as a figure skater pulling their arms in.

As it spins it flattens into a protoplanetary disk. The central region reaches high temperatures and high densities, the material igniting to form a protostar. Once the core reaches the temperature for hydrogen fusion to begin, a star is born. The disk surrounding the protostar starts to form planets as the little grains of dust and ice begin to stick to one another by accretion, and as the small pieces gather, they gain a bigger mass. Once they reach a certain size, "planetesimals" formed and these collided to make planets. Depending on the temperature of the disk at this position,

planets can be rocky-forming on the hot inner areas of the disk, or become huge gas planets accreting mantles of gas around them beyond the "frost line".

The two end states of stars much more massive than the Sun are neutron stars and black holes.

If a star with a significantly larger mass than the Sun runs out of fuel and experiences a supernova the remaining core collapses into one of two unusual objects:

1. *Neutron Stars*: If the remaining core has a mass of approximately between 1.4 and 3 Suns it collapses into a sphere of just 20km across. Such objects are so dense that their protons and electrons merge together to form neutrons, a mere teaspoonful of a neutron star would have a mass measured in billions of tonnes. Neutron stars are kept from collapsing further by neutron degeneracy pressure and are observed as pulsars; stars emitting jets of radiation as they rotate rapidly.
2. *Black Holes*: If the remaining core of a supernova has a mass of greater than 3 Suns nothing is thought to be able to hold back its collapse to a mathematical point of infinite density known as a singularity. The immense gravity generated by this object warps spacetime around it and forms an area with an escape velocity greater than the speed of light, which is known as the event horizon. Any object crossing the event horizon may not re-enter the visible universe.